Converse of theorem 9

The chords of a circle which are equidistant from the centre are equal.

4.5.3 Angle Subtended by an Arc of a Circle

Activity - 6

Procedure :

1. Draw three circles of any radius with centre O on a chart paper.
2. From these circles, cut a semi-circle, a minor segment and a major segment.
3. Consider three points on these segment and name them as A, B and C.

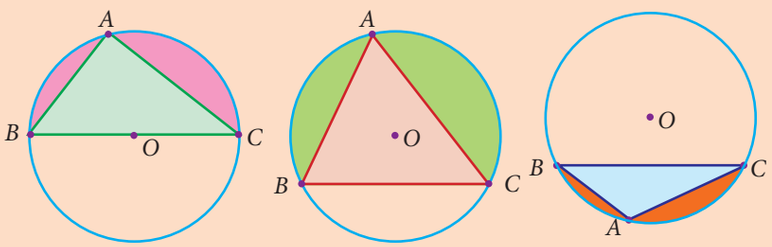


Fig. 4.63

4. (iv) Cut the triangles and paste it on the graph sheet so that the point A coincides with the origin as shown in the figure.

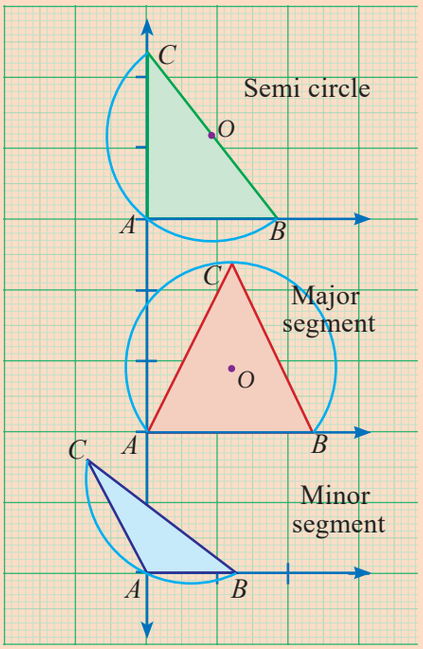


Fig. 4.64

Observation :

(i) Angle in a Semi-Circle is _____ angle.

(ii) Angle in a major segment is _____ angle.

(iii) Angle in a minor segment is _____ angle.

Now we are going to verify the relationship between the angle subtended by an arc at the centre and the angle subtended on the circumference.

4.5.4 Angle at the Centre and the Circumference

Let us consider any circle with centre O. Now place the points A, B and C on the circumference.

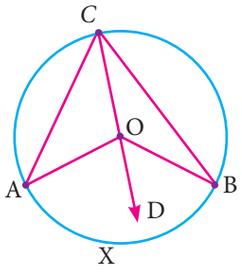


Fig. 4.65

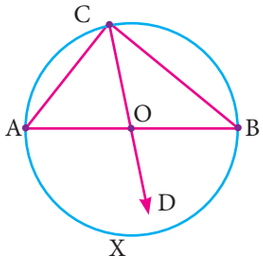


Fig. 4.66

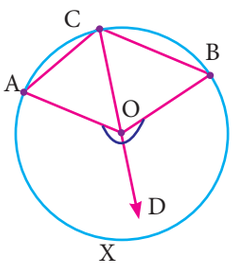


Fig. 4.67

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